

Prerak Bhandari

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EDUCATION

M.S. Computer Science - North Carolina State University, Raleigh <i>Coursework: Design and Analysis of Algorithms, Computer Networks, DevOps, OOD.</i>	August 2024 - May 2026 GPA: 4.00/4.00
B.E. Computer Science - R. N. S. Institute of Technology, India <i>Coursework: AI/ML, Operating Systems, Distributed Systems.</i>	July 2018 - August 2022 GPA: 9.02/10.00

TECHNICAL SKILLS

Programming Languages: Go, Python, Java, C, JavaScript, TypeScript, Ruby, SQL, HTML, CSS, PHP.
Frameworks: Spring Boot, React.js, Node.js, Django, Flask, FastAPI, Ruby on Rails
Databases: MongoDB, MySQL, PostgreSQL, Redis, Elasticsearch, DynamoDB.
Tools: AWS (EKS, S3, EC2, Lambda, ELB, SQS), Azure, Docker, Kubernetes, Ansible, Git, RabbitMQ, Jira, Jenkins

WORK EXPERIENCE

Graduate Student Developer - OSS(Expertiza) <i>North Carolina State University</i>	January 2026 - Present <i>Raleigh, NC</i>
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- Designed and engineered backend features for **Expertiza OSS (Ruby on Rails, TypeScript)** supporting scalable peer review workflows, rubric-based grading, and submission evaluation pipelines.
- Refactored legacy grading models into a reusable hierarchy, reducing code duplication by **30%**, and **enhanced submission response APIs** to improve peer review workflow reliability and backend testability.

Software Engineer Intern <i>Tailon Labs</i>	May 2025 - August 2025 <i>Remote (USA)</i>
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- Developed a **full-stack Real Estate Management** platform using **React.js, Node.js, TypeScript, and PostgreSQL**, enabling realtors to manage property listings at scale, and **improving operational efficiency by 35%**.
- Optimized **RESTful API** architecture with asynchronous I/O, connection pooling, and Redis-based caching, achieved **3 times throughput** and **reduced p95 latency from 280 ms to 85 ms** under load.

Software Engineer <i>Kalegra Inc.</i>	April 2022 - July 2024 <i>Bangalore, India</i>
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- Owned and scaled **SMS pipelines** to handle **millions of messages**; overhauled **Sender ID, template, and content APIs**, improved message delivery by **20%** and campaign performance.
- Engineered **multi-channel messaging (Failover)** across SMS, WhatsApp, RCS, Voice, Email using AWS SQS based failover routing to achieve **99.9% reliability** and reduced delivery failures by **40%**.
- Built **WhatsApp capabilities** with **Meta Graph API** (Tech Provider, Account Sharing), automated billing and eliminated billing errors.
- Led implementation of custom domain routing and analytics tracking for the URL Shortener platform across a **3 engineer team**, powering enterprise messaging workloads reaching **5M+ end users**.
- Migrated **150+ APIs** from PHP to **Go** using **gRPC**, improved **latency by 97.5%** and scaled **throughput from 4000 to 15,000 requests per second** while moving the platform from monolith to **microservices**.
- Developed real-time campaign tools including an **Excel SMS Add-in** which improved workflow efficiency by **25%**, and contributed to an **18%** revenue increase.

PROJECTS

Sword URL Shortener (swrd.shop) (Frontend Source Code) (Backend Source Code) - Built a 99.9% uptime production-grade URL Shortener using Java (Spring Boot), PostgreSQL, and React.js; scaled to handle 1000+ RPS with JWT authentication, real-time analytics, and RabbitMQ-based background processing. - Built an end-to-end CI/CD pipeline running automated tests and linting, packaging services into Docker images, and deploying to AWS EC2 via Ansible with controlled release flows from develop to main.
Smart Inventory AI Bot (Source Code) - Developed a cross-platform, LLM-integrated mobile app (React Native, FastAPI, Google Gemini AI) that connects to Google Drive CSVs for real-time inventory, customer, and pricing queries. - Built a structured query analysis and conversational response pipeline, reducing query-to-answer latency by 40%. - Integrated Google Drive API with caching (pandas DataFrames) for live data insights across 5000+ records.

PUBLICATION

- Prerak Bhandari**, Dr. E. F. Gehringer “*LLM Conversation Transcripts in Software Engineering: Redefining Homework in the Age of Generative AI.*” 2026 American Society for Engineering Education (ASEE) Annual Conference and Exhibition, Charlotte, NC